

Denver_Health_Crime

2026-07-29

Denver Crime Rates & Health Facility Access

- This project is understanding crimes rates in the Denver area and their proximity to health facilities.
 - Specifically, this data is looking at violent crimes, substance crimes, and total crimes combined with mental health facilities, substance treatment facilities, and total health centers. All of this is split up into the different zip codes of the Denver metro area.
- This project uses data from the Colorado Information Marketplace and the Open Data Catalog.

Descriptive Statistics

- First, the data needs to be brought into R from the CSV file that we created using SQL and Python.

```
df <- read.csv('clean_crimes.csv')
```

- Next, looking at the total amount of crimes in the Denver area and looking at the zip code with the most crimes overall.

```
total_crime <- sum(df$total_crimes)
```

```
total_crime
```

```
## [1] 213223
```

```
total_crime_table <- aggregate(df$total_crimes, list(zip_codes = df$zip_code), FUN = 'mean')
```

```
total_crime_table <- subset(total_crime_table, x > 0)
```

```
total_crime_table
```

```
##      zip_codes      x
## 27      80123  1580
## 34      80202 11344
## 35      80203 11117
## 36      80204 15802
## 37      80205  9245
## 38      80206 14687
## 39      80207  6872
## 41      80209  4299
## 42      80210  7811
## 43      80211  8242
```

```
## 44      80212  2874
## 47      80216 16886
## 48      80218   819
## 49      80219 15032
## 50      80220 12027
## 51      80221   215
## 52      80222   402
## 53      80223   704
## 54      80224   553
## 56      80227   147
## 59      80230   142
## 65      80236   872
## 66      80237   939
## 67      80238 12081
## 68      80239   524
## 69      80247   391
## 70      80249   876
```

- There is only 27 zip codes that have crimes greater than 0 which is included in the table above along with the total crimes in the Denver Metro Area.
- I'd like to do the same for the health centers now.

```
total_centers <- sum(df$total_health_centers)

total_centers
```

```
## [1] 506
```

```
total_centers_table <- aggregate(df$total_health_centers, list(zip_codes = df$zip_code), FUN = 'mean')
total_centers_table <- subset(total_centers_table, x > 0)

total_centers_table
```

```
##      zip_codes  x
## 1      80002   1
## 2      80003   3
## 3      80010   5
## 4      80011  10
## 5      80012   3
## 6      80013   1
## 7      80014   8
## 8      80015   1
## 9      80020   2
## 10     80021   3
## 11     80022   6
## 12     80026   3
## 13     80027   2
## 14     80030   4
## 15     80031   2
## 16     80033  10
## 17     80045   3
## 18     80104   1
```

## 19	80107	2
## 20	80108	1
## 21	80110	2
## 22	80111	3
## 23	80112	9
## 24	80113	1
## 25	80120	6
## 26	80121	1
## 27	80123	1
## 28	80124	1
## 29	80128	4
## 30	80132	2
## 31	80133	1
## 32	80134	6
## 33	80138	1
## 35	80203	2
## 36	80204	7
## 37	80205	7
## 38	80206	4
## 40	80208	1
## 41	80209	1
## 42	80210	3
## 43	80211	2
## 44	80212	2
## 45	80214	3
## 46	80215	3
## 47	80216	1
## 48	80218	1
## 49	80219	5
## 50	80220	4
## 51	80221	1
## 52	80222	6
## 53	80223	2
## 54	80224	1
## 55	80226	5
## 56	80227	2
## 57	80228	1
## 58	80229	1
## 60	80231	5
## 61	80232	1
## 62	80233	4
## 63	80234	3
## 64	80235	1
## 65	80236	8
## 66	80237	1
## 68	80239	2
## 71	80260	5
## 72	80301	2
## 73	80302	3
## 74	80303	2
## 75	80304	6
## 76	80305	1
## 77	80401	2
## 78	80421	1

## 79	80422	1
## 80	80433	1
## 81	80435	1
## 82	80439	1
## 83	80440	1
## 84	80443	3
## 85	80446	3
## 86	80452	2
## 87	80461	2
## 88	80477	3
## 89	80480	2
## 90	80482	1
## 91	80487	1
## 92	80501	11
## 93	80504	1
## 94	80513	1
## 95	80514	2
## 96	80517	2
## 97	80521	1
## 98	80524	5
## 99	80525	5
## 100	80526	4
## 101	80528	1
## 102	80530	2
## 103	80534	2
## 104	80537	5
## 105	80538	2
## 106	80550	1
## 107	80601	5
## 108	80621	2
## 109	80631	11
## 110	80634	3
## 111	80701	6
## 112	80720	2
## 113	80734	2
## 114	80737	2
## 115	80751	3
## 116	80758	2
## 117	80759	2
## 118	80807	2
## 119	80810	1
## 120	80813	1
## 121	80817	2
## 122	80828	1
## 123	80840	1
## 124	80863	1
## 125	80866	1
## 126	80903	4
## 127	80904	3
## 128	80905	2
## 129	80906	5
## 130	80907	3
## 131	80909	5
## 132	80910	2

## 133	80915	1
## 134	80916	1
## 135	80917	1
## 136	80918	7
## 137	80920	1
## 138	80923	1
## 139	81001	6
## 140	81003	4
## 141	81004	2
## 142	81005	1
## 143	81008	4
## 144	81050	5
## 145	81052	3
## 146	81054	3
## 147	81063	2
## 148	81067	1
## 149	81073	2
## 150	81082	2
## 151	81089	2
## 152	81101	5
## 153	81120	2
## 154	81125	2
## 155	81135	1
## 156	81137	2
## 157	81140	2
## 158	81144	2
## 159	81147	2
## 160	81152	2
## 161	81201	2
## 162	81211	1
## 163	81212	4
## 164	81224	1
## 165	81230	3
## 166	81301	5
## 167	81303	1
## 168	81321	3
## 169	81401	2
## 170	81402	1
## 171	81416	3
## 172	81423	2
## 173	81428	1
## 174	81435	2
## 175	81501	14
## 176	81503	1
## 177	81504	1
## 178	81601	6
## 179	81611	2
## 180	81623	4
## 181	81625	5
## 182	81631	3
## 183	81632	1
## 184	81641	1
## 185	81648	2
## 186	81650	2

```
## 187      81657  2
```

- There are a small number of total health centers; however, there is at least one health center in I believe all the different neighborhoods.

Correlations

- For more main analyses I'd like to run a correlation between the number of violent crimes and the number of mental health centers.

```
correlation_violent <- cor(df$violent_crimes,df$mental_health_centers, method = 'pearson')
correlation_violent
```

```
## [1] -0.06391106
```

- The value of the correlation is -0.064 which is a non-existent relationship between these two variables.
- I'd like to see the correlation of substance crimes and the number of substance treatment centers next.

```
correlation_substance <- cor(df$substance_crimes,df$substance_treatment_centers, method = 'pearson')
correlation_substance
```

```
## [1] -0.02386959
```

- The r value here is -0.024, which is another non-existent relationship between these two variables.

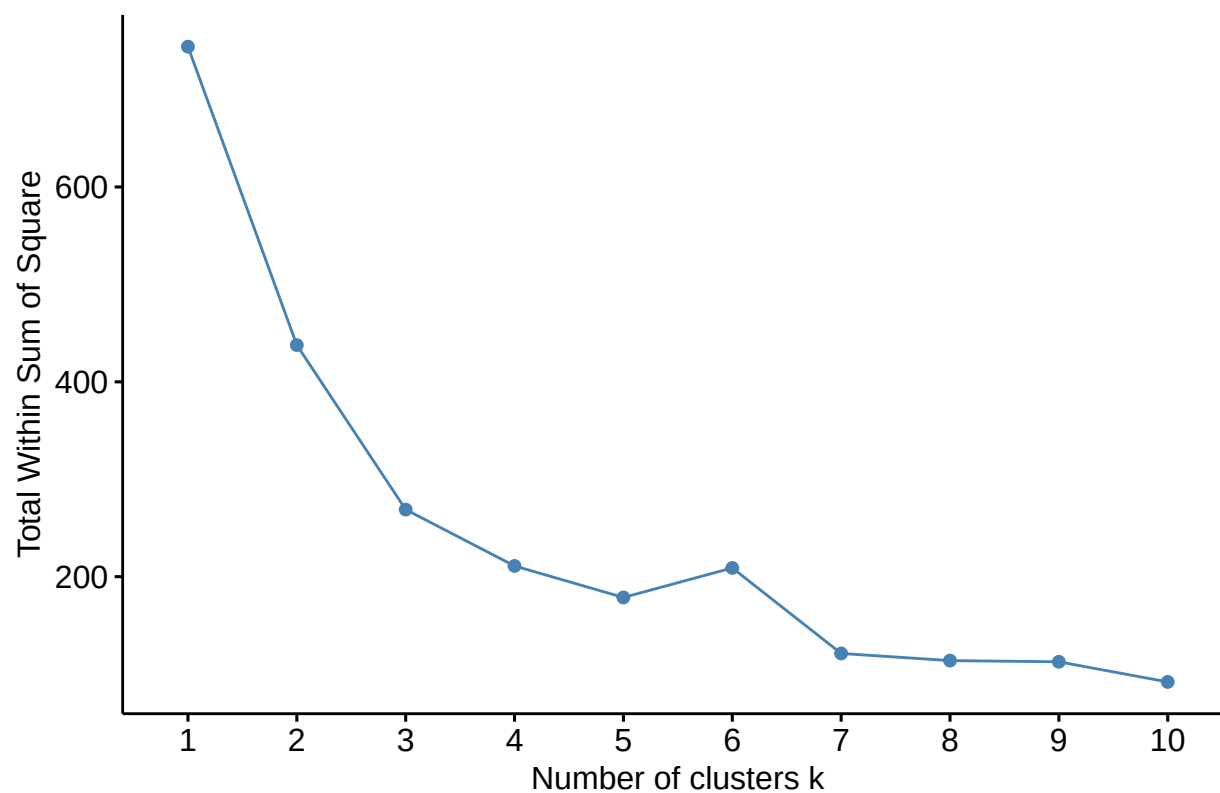
K-mean Clustering

- I want to check out some unsupervised machine learning with K-means clustering.

```
cluster_data <- df %>%
  select(total_crimes, violent_crimes, total_health_centers, mental_health_centers) %>%
  scale()

fviz_nbclust(cluster_data, kmeans, method = "wss") +
  labs(title = "Elbow Method for Optimal Clusters")
```

Elbow Method for Optimal Clusters



- Looks like the optimal number of clusters is 9. I'm planning on looking at 3 and 5 just to see the graphs.

```
set.seed(303)
kmeans_result <- kmeans(cluster_data, centers = 3, nstart = 25)

df$cluster_profile <- as.factor(kmeans_result$cluster)

fviz_cluster(kmeans_result, data = cluster_data,
  geom = "point",
  ellipse.type = "convex",
  ggtheme = theme_minimal(),
  main = "Zip Code Cluster Profiles")
```



- After looking at the graphs I'd say 3 is most ideal in connection with this project since three categories exist and the clustering works well.

```
cluster_summary <- df %>%
  group_by(cluster_profile) %>%
  summarize(
    avg_crime = mean(total_crimes, na.rm = TRUE),
    avg_violent_crime = mean(violent_crimes, na.rm = TRUE),
    avg_health_clinics = mean(total_health_centers, na.rm = TRUE),
    avg_mental_health = mean(mental_health_centers, na.rm = TRUE),
    zip_count = n()
  )

print(cluster_summary)
```

```
## # A tibble: 3 x 6
##   cluster_profile avg_crime avg_violent_crime avg_health_clinics
##   <fct>          <dbl>          <dbl>          <dbl>
## 1 1             10936.           654.           2.94
## 2 2              252.           13.9           2.13
## 3 3              0            0            7.11
## # i 2 more variables: avg_mental_health <dbl>, zip_count <int>
```

- Now assign the personas.


```
df <- df %>%  
  mutate(  
    persona = case_when(  
      cluster_profile == 1 ~ "Intervention Priority",  
      cluster_profile == 2 ~ "Baseline",  
      cluster_profile == 3 ~ "Resource Hubs",  
      TRUE ~ "Other"  
    )  
  )
```

- Now send the csv back out

```
write.csv(df, "denver_zip_personas.csv", row.names = FALSE)
```